

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
**Department of Artificial Intelligence and
Data Science**

USN:

Week #: 02

Semester:

Section:

Date:

Instructions:

- **Implement the following programs using python language.**

Programs:

1. Create a function `sum()` and a function `multiply()` that takes a list of numbers and displays the sum of all the numbers if `sum()` is called and displays multiplication of all the numbers if the `multiply()` is called. For example, `sum([1, 2, 3, 4])` should return 10, and `multiply([1, 2, 3, 4])` should return 24.
2. Create a `displayMinMax()` function that takes an Array and displays the maximum and minimum number from the array. Create a `displaySecondMax()` to display the second largest number from the array without sorting.
3. Write a python function `linearSearch()` to read an array and search for the key element. Display the appropriate messages. Use the recursive function.
4. Write a python function `binarySearch()` to read a sorted array and search for the key element. Display the appropriate messages.
5. Create a `displayFibonacci()` function that displays the Fibonacci series between 0 to 100.
6. Create a `findQuadraticRoots()` function that finds the roots of a quadratic equation. Display appropriate messages.
7. Write a python program to create a list with all the subject names of the 4th semester and perform the following operations.
 - Display the list using for loop.
 - Display 2nd and 5th element of the list.
 - Display first 4 elements of the list using the range of indexes.
 - Display last 4 elements of the list using the range of negative indexes.
 - Display if "Python Programming Lab" is available in the List or not.
 - Demonstrate the working of `append()` and `insert()` function.
 - Demonstrate the working of `remove()` and `pop()` function.



- Demonstrate the working of extend() function by adding subject codes.
- Display the list in ascending and descending order.

EVALUATION			
Program	Remarks	Marks	Faculty Signature
Program - 1			
Program – 2			
Program – 3			
Program – 4			
Program – 5			
Program – 6			
Program – 7			